Department of Software Engineering

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SettleIn – A MERN Stack Based Smart Web Application with AI Assistance for Supporting International Students

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DECLARATION

I certify that project title _SettleIn – A MERN Stack Based Smart Web Application with AI Assistance and Push Notifications for Supporting International Students is under my supervision with students of M ZOHAIB, M SAAD, UMM-E-HABIBA, Faculty of Computing & Information Technology, University of Gujrat, Pakistan, worked under my supervision.

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TABLE OF CONTENTS

FINAL YEAR PROJECT PROPOSAL 4

1.1 INTRODUCTION..... 4
1.2. PROJECT TITLE:..... 4
1.3. PROJECT OVERVIEW STATEMENT:..... 4
1.4. PROJECT GOALS & OBJECTIVES:..... 6
1.5. HIGH-LEVEL SYSTEM COMPONENTS: 6
1.6. LIST OF OPTIONAL FUNCTIONAL UNITS:..... 6
1.7. EXCLUSIONS: 7
1.8. APPLICATION ARCHITECTURE:..... 7
1.9. GANTT CHART:..... 7
1.10. HARDWARE AND SOFTWARE SPECIFICATION: 8
1.11. TOOLS AND TECHNOLOGIES USED WITH REASONING: 8

Final Year Project Proposal

1.1 Introduction

International students face various challenges when they arrive in a new country, such as finding accommodation, managing documents, understanding local culture, and handling emergencies. The absence of a proper support system often causes stress and affects their academic as well as personal life. To address these issues, this project proposes SettleIn, *a MERN stack based smart web application. It provides features like a first-week checklist, housing and roommate finder, emergency contacts, student deals, and a community forum. The system also integrates an AI chatbot for guidance and push notifications for timely alerts, ensuring a smoother and more supportive experience for international students.*

1.2. Project Title:

The title of this project is **"SettleIn – A MERN Stack Based Smart Web Application with AI Assistance and Push Notifications for International Students."** This title clearly reflects the purpose and scope of the system, which is to provide a digital support platform for international students. The most important elements of the title are mentioned first, including the technology stack (MERN), the nature of the system (smart web application), and the core features (AI assistance and push notifications). The title is concise yet descriptive, giving a quick picture of what the project aims to achieve without being vague or overly complex.

1.3. Project Overview statement:

International students often face serious challenges when moving to a new country, including accommodation issues, cultural adjustment, managing documents, finding jobs, and handling emergencies. These problems can create stress and negatively affect both academic and personal life. To address these concerns, this project proposes **SettleIn**, a MERN stack based smart web application that provides international students with a reliable support system.

The application offers multiple features such as a first-week checklist, housing and roommate finder, document manager, emergency contacts, community forum, and student deals. An AI chatbot is integrated to provide instant guidance, while push notifications ensure timely alerts and reminders for important tasks. By combining these features, SettleIn aims to make the adjustment process easier, less stressful, and more organized for students. The system is designed to be scalable, user-friendly, and practical, ensuring that it can be adopted by educational institutions or student support organizations to improve the overall experience of international students.

Project Overview Statement Template

Project Title: SettleIn – A MERN Stack Based Smart Web Application with AI Assistance for Supporting International Students

Project Manager: MIRZA AHSAN ULLAH

Project Members:

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Project Goal: The primary goal of this project is to develop a reliable digital support system that helps international students overcome the difficulties they face when settling in a new country. The project aims to make their adjustment process smoother, more organized, and less stressful by providing a single web-based platform that combines academic, social, and personal support. By using modern technologies such as the MERN stack, AI chatbot, and push notifications, the project also seeks to deliver a scalable, accessible, and user-friendly solution.

Objectives:

Sr.#	
1	Develop a MERN-based web application with secure authentication.
2	Provide a first-week checklist to guide international students.
3	Build a document manager with reminders for important records.
4	Implement a housing and roommate finder with AI-based suggestions.
5	Integrate an AI chatbot (SettleBot) for real-time guidance and queries.
6	Add emergency help features with local contacts and safety resources.

Project Success criteria: The SettleIn project will be considered successful if the core modules are fully implemented and work as intended. Success will be measured by:

- A functional MERN-based web application with secure login.
- Working modules: checklist, housing finder, AI chatbot, community forum, and emergency help.
- Push notifications delivering timely alerts.
- User-friendly interface accessible on modern browsers.
- Successful deployment on cloud platforms.
- Positive feedback from test users.

Assumptions, Risks and Obstacles: It is assumed that international students will have internet access, basic digital literacy, and provide accurate information while institutions may support adoption of the platform. Major risks include potential inaccuracies from the AI chatbot, security vulnerabilities in handling sensitive student data, and dependency on third-party services like Firebase or MongoDB, which may face limitations or downtime. Key obstacles involve budget and time constraints, technical challenges in integrating multiple services, ensuring scalability and cross-platform compatibility, and encouraging students to adopt the system actively.

Organization Address (if any): None

Type of project:	Development
Target End users:	International Students
Development Technology:	Structured
Platform:	Web based
Approved By:	
Date:	

1.4. Project Goals & Objectives:

The primary goal of this project is to develop a reliable digital support system that helps international students overcome the difficulties they face when settling in a new country. The project aims to make their adjustment process smoother, more organized, and less stressful by providing a single web-based platform that combines academic, social, and personal support. By using modern technologies such as the MERN stack, AI chatbot, and push notifications, the project also seeks to deliver a scalable, accessible, and user-friendly solution.

To achieve these goals, the project sets specific objectives. These include developing a MERN-based web application with secure authentication, a first-week checklist, and a document manager with reminders. The system will also provide a housing and roommate finder, an AI chatbot for real-time guidance, emergency help features, and a community forum for peer support. Furthermore, push notifications will ensure that students receive timely alerts, while additional modules like student deals, part-time job alerts, and budget tracking will enhance the overall experience. Together, these objectives form a practical roadmap for meeting the broader goals of the project.

1.5. High-level system components:

The SettleIn system is composed of several high-level components that work together to provide a complete solution for international students. At the core, the **user authentication and profile management** module ensures secure access and personalized experiences. The **first-week checklist and document manager** component guides students through essential tasks and helps them store and manage important documents with reminders. The **housing and roommate finder** provides search filters and AI-based suggestions to make living arrangements easier. An **AI chatbot (SettleBot)** serves as an intelligent assistant, answering queries and providing quick guidance. The **emergency help and local guide** module gives students access to nearby services and safety contacts. The **community forum** supports peer-to-peer interaction, while the **student deals, jobs, and budget tracker** component helps students discover discounts, job opportunities, and manage finances. Finally, the **push notification system** ensures that timely alerts and reminders are delivered to users. Together, these components create a unified platform that fulfills the core mission of supporting international students in their transition.

1.6. List of optional functional units:

In addition to the core modules of SettleIn, there are several optional functional units that may be included if time, technology, or resources allow. One such feature is **offline support**, where students could access essential information like emergency contacts and

checklists even without an internet connection through Progressive Web App capabilities. Another optional unit is an **advanced financial management system**, which would allow students to create detailed budgets, receive AI-driven financial advice, and track expenses more effectively. Similarly, the platform could include **mental health and counseling support links**, giving students direct access to professional resources when needed. Other optional features may involve **multilingual support for a wider audience**, **data analytics for institutions** to understand student needs better, and **enhanced security auditing tools** for stricter compliance. While these units are not critical for the initial version of the system, their inclusion would significantly improve usability, accessibility, and overall quality of the platform in the future.

1.7. Exclusions:

The scope of this project is limited to providing international students with digital support through a MERN-based web application. Therefore, certain features are excluded from the system. The platform will not handle **financial transactions** such as tuition fee payments or online banking, as these require integration with secure financial systems beyond the scope of this project. It will also exclude **government visa or legal processing services**, since these involve official procedures that cannot be automated reliably within the application. Similarly, the system will not function as a full **Learning Management System (LMS)** for academic courses, as its primary focus is on student support outside the classroom. These exclusions ensure that the project remains practical, focused, and achievable within the available time and resources.

1.8. Application Architecture:

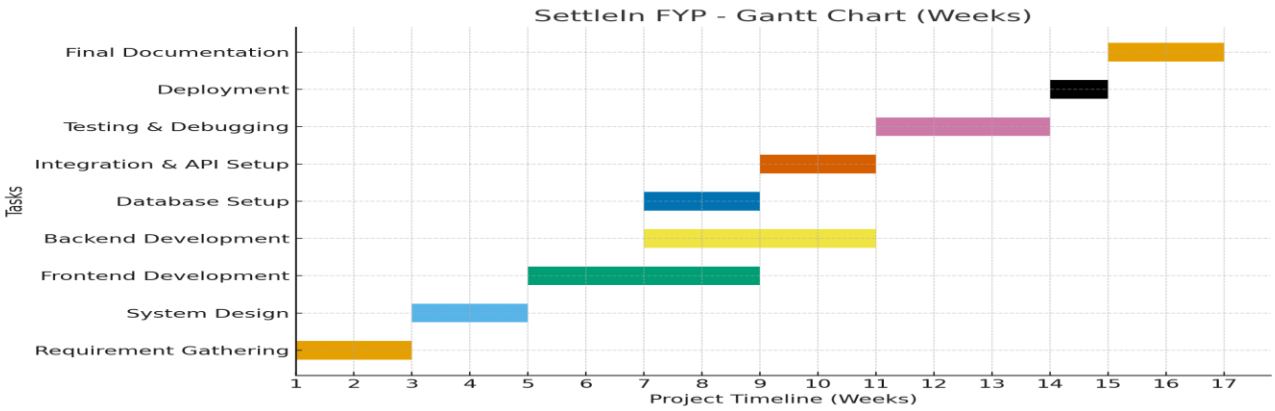
The architecture of the SettleIn system is based on a three-tier model, consisting of the **frontend, backend, and database layers**. The frontend is developed using **React.js**, which provides a responsive and user-friendly interface for students to interact with the system. The backend is powered by **Node.js and Express.js**, which handle business logic, API integrations, and secure communication between different components. The database layer uses **MongoDB Atlas**, a cloud-based NoSQL database that ensures scalability and efficient storage of user data, documents, and application content. In addition, the system integrates with external services such as the **OpenAI API** for the chatbot, **Firebase Cloud Messaging** for push notifications, and **Google APIs** for maps and translation. This layered architecture ensures modularity, scalability, and ease of maintenance, while also supporting real-time communication and secure access for users.

1.9. Gantt chart:

The development of SettleIn will be carried out in multiple phases following an iterative approach. The project begins with **requirement gathering and analysis**, which helps in defining the functional and non-functional requirements of the system. This is followed by **system design**, where both the architecture and user interface are planned. After the design phase, the team will move to **frontend and backend development**, along with **database setup** and API integrations. Once the core modules are built, the **integration**

phase will ensure smooth communication between different components.

The next stage involves **testing and debugging**, where unit testing, integration testing, and system testing will be performed to ensure quality and reliability. Finally, the project will proceed to **deployment** on cloud platforms and **final documentation**, which includes preparing the user manual, technical documentation, and project report.



1.10. Hardware and Software Specification:

The development of the SettleIn system requires standard hardware and software resources. On the hardware side, a laptop or desktop computer with at least an **Intel i5 processor (or equivalent), 8GB RAM, and 256GB storage** is sufficient for development and testing purposes. For software, the project will be developed on **Windows or Linux operating systems** with **Node.js** and **npm** for backend and package management, **MongoDB Atlas** for cloud database services, and **Visual Studio Code (VS Code)** as the main development environment. Additional utilities include **Postman** for API testing, **Git and GitHub** for version control, and **Vercel/Render** for deployment. The system will run in any modern web browser such as **Google Chrome or Microsoft Edge**, ensuring accessibility for end users without the need for high-end hardware.

1.11. Tools and technologies used with reasoning:

For the development of SettleIn, a MERN stack–based web application, a combination of modern tools and technologies is selected to ensure scalability, performance, and ease of use.

- **Frontend (React.js with Tailwind/Material UI):** React.js is chosen because it supports building responsive and dynamic interfaces efficiently. Tailwind CSS and Material UI help create a clean, user-friendly design with minimal effort.
- **Backend (Node.js with Express.js):** Node.js allows handling multiple concurrent requests efficiently, making it ideal for real-time features like chatbots and notifications. Express.js simplifies API development and backend routing.
- **Database (MongoDB Atlas):** MongoDB, a NoSQL cloud database, provides

flexibility in handling unstructured data such as user profiles, checklists, housing details, and forum posts. Its scalability supports large datasets as the user base grows.

- **AI Integration (OpenAI API / Rasa):** An AI chatbot (SettleBot) is powered by OpenAI API to answer queries and provide intelligent support. This improves the overall student experience and reduces manual support requirements.
- **Push Notifications (Firebase Cloud Messaging):** Firebase is used to implement browser-based push notifications, ensuring that users receive timely alerts such as checklist reminders, housing updates, and emergency alerts.
- **Authentication (JWT):** JSON Web Tokens are used to secure user authentication and maintain session management.
- **Development Tools:**
    - **Visual Studio Code (VS Code):** Main IDE for coding.
    - **Postman:** For testing REST APIs during development.
    - **Git & GitHub:** For version control and team collaboration.
    - **Jest & Cypress:** For unit and end-to-end testing in iterative development cycles.
- **Deployment Platforms:**
    - **Frontend:** Vercel/Netlify for fast, reliable hosting.
    - **Backend:** Render/Heroku for server deployment.
    - **Database:** MongoDB Atlas (cloud hosting).
- **Host / Target Platforms:** The development platform will be Windows/Linux laptops, while the target platform will be **web browsers** (Google Chrome, Firefox, Edge), making the system easily accessible without requiring high-end devices.
- **Programming Languages:** JavaScript (ES6+) is used for both frontend and backend, providing consistency across the project.
- **Constraints:**
    - The project follows an **iterative development process**, so automated testing tools like Jest and Cypress are required to ensure quality in repeated testing cycles.
    - As the project is developed by a small team, tools are selected for their simplicity, cost-effectiveness, and open-source availability.
    - Since the organization is not physically distributed, collaboration tools like GitHub are sufficient without requiring heavy enterprise-level project management software.
    - Budget and time constraints encourage the use of free or student-supported services such as MongoDB Atlas free tier, GitHub Student Pack, and Firebase free plan.

By carefully choosing these tools and technologies, the project ensures a balance between functionality, cost, scalability, and ease of maintenance, making it suitable for both academic and practical deployment.